



Adaptive area-wide management for suppressing spruce budworm outbreaks: effectiveness of the early intervention strategy

Rob C. Johns¹ · S. Edwards¹ · L. Amos-Binks³ · J. J. Bowden⁴ · D. Carleton³ · C. Edge¹ · C. Hennigar¹ · P. James⁵ · D. MacLean⁶ · V. Martel² · E. R. D. Moise⁴ · B. Morin¹ · A. Morrison³ · E. Owens¹ · M. Stastny¹ · J. Régnière²

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Abstract

In eastern North America, two main insecticide-based strategies are used to manage spruce budworm (*Choristoneura fumiferana*) outbreaks. The traditional ‘Foliage Protection Strategy’ (FPS) focuses on temporarily reducing defoliation and tree mortality in high-value stands, allowing outbreaks to progress naturally in surrounding areas. The ‘Early Intervention Strategy’ (EIS) is an adaptive, area-wide management approach that targets low-density “hotspots” to prevent outbreak growth and spread. We report findings from a 10-year case study (2015–2024) in Atlantic Canada assessing EIS effectiveness. EIS treatments consistently reduced population growth rates, resulting in a 12–67% reduction in hotspot areas compared to untreated areas. Crucially, this suppression curbed the rise and spread of the outbreak across northern New Brunswick, with negligible defoliation observed throughout the study period. By comparison, in a neighboring Quebec study area contemporaneously managed with FPS, the outbreak advanced normally, causing moderate to severe defoliation over ~1.7 million hectares by 2024. Moreover, EIS required ~80% less insecticide, with 66% of treated areas receiving treatment in only a single year, compared with just 15% under FPS. Our findings provide robust evidence for the effectiveness of EIS when implemented under suitable conditions, highlighting its potential as a proactive outbreak management approach.

Keywords Area-wide management · Early intervention strategy · Proactive · Integrated pest management · Foliage protection strategy

Introduction

Insect outbreaks place significant economic and ecological strain on North American forests, disrupting timber supply (Hennigar et al. 2013), ecosystem services (Schowalter et al. 2018), forest structure (Kneeshaw et al. 2022), and carbon budgets (Liebhold et al. 2016; Liu et al. 2020). These effects are especially severe for species that undergo regional outbreaks, which often cover areas far exceeding those typically affected by fire or harvesting (Sturtevant et al. 2015). In response, extensive research has focused on developing and testing strategies for managing forest insect pests. While small-scale mesocosm experiments provide useful insights into the localized and short-term efficacy of certain tactics, they frequently fail to capture the complexity of large-scale management (Schindler 1998). Therefore, a more accurate evaluation of management effectiveness comes from operational-scale studies, where outcomes reflect the full range of ecological, logistical, and sociopolitical influences.

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✉ Rob C. Johns
rob.johns@nrcan-rncan.gc.ca

- ¹ Natural Resources Canada, Canadian Forest Service - Atlantic Forestry Centre, Fredericton, NB, Canada
- ² Natural Resources Canada, Canadian Forest Service - Laurentian Forestry Centre, Quebec City, QC, Canada
- ³ New Brunswick Department of Natural Resources - Forest Planning and Stewardship Branch, Fredericton, NB, Canada
- ⁴ Natural Resources Canada, Canadian Forest Service - Atlantic Forestry Centre, Cornerbrook, NF, Canada
- ⁵ Graduate Department of Forestry, University of Toronto, Toronto, ON, Canada
- ⁶ Faculty of Forestry and Environmental Management, University of New Brunswick, Fredericton, NB, Canada

Among the approximately 106 defoliating insect species known to outbreak in Canadian forests (Johns et al. 2016), few are as destructive as the spruce budworm, *Choristoneura fumiferana* Clem. (Lepidoptera: Tortricidae). This native insect is the primary defoliator of balsam fir (*Abies balsamea* [Mills] B.S.P.) and spruce (*Picea* spp.), with a transcontinental range spanning the North American boreal forest from Alaska to Newfoundland. Spruce budworm outbreaks have occurred approximately every 35 to 40 years for centuries (Boulanger et al. 2012; Sturtevant et al. 2015), with subfossil records suggesting outbreaks dating back to the early Holocene (Simard et al. 2006). In the twentieth century alone, three major outbreaks swept through eastern Canada, the last and most severe lasting from 1967 to 1993 (Pureswaran et al. 2016). That outbreak was the most extensive on record, affecting all eastern Canadian provinces and the northeastern United States, and resulting in over 50 million hectares of defoliation. Balsam fir is especially vulnerable compared to spruce, suffering more severe impacts from repeated defoliation (Hennigar et al. 2008). Mortality in balsam fir typically begins 5–7 years after the onset of sustained defoliation (MacLean 1980), whereas black spruce rarely experiences significant mortality. During a typical local outbreak, which can last around a decade, balsam fir may suffer up to 90% reduction in growth with mortality rates exceeding 85% in some stands (MacLean 1980; Ostaff and MacLean 1989). These impacts have far-reaching implications for forest succession (Bouchard et al. 2006), carbon dynamics (Liu et al. 2022), and susceptibility to secondary disturbances such as fire (James et al. 2017) and herbivory by other species (Leroux et al. 2021).

The economic consequences of spruce budworm outbreaks are similarly profound. During the last outbreak in Quebec, an estimated 238 million cubic meters of softwood were lost, equating to commercial losses of approximately \$12.5 billion (Lévesque et al. 2010). In New Brunswick—where roughly 85% of the forest is composed of susceptible species (Hennigar et al. 2013)—Chang et al. (2012a, b) estimated that an uncontrolled future outbreak of moderate to severe intensity could lead to economic losses ranging from \$3.3 to \$4.7 billion CAD over a 30-year period.

Given the widespread impacts of spruce budworm, substantial theoretical and applied research has focused on developing effective strategies to mitigate its damage (Sturtevant et al. 2015; Holmes and MacQuarrie et al. 2016; Pureswaran et al. 2016). Prior to outbreaks, silvicultural approaches—such as altering forest composition—can reduce the abundance of highly susceptible species like balsam fir (Hennigar et al. 2008), or increase the proportion of hardwoods to support more resilient natural enemy communities (Legault and James 2018; Zhang et al. 2020), thus potentially reducing the synchrony, frequency, and intensity of outbreaks (Robert et al. 2012). In the later stages of an

outbreak, adaptive harvesting strategies, including the timely harvest of affected trees, can help recover economic value before decay renders them unsuitable for lumber or pulp (the “salvage” strategy; Miller and Rusnock 1993; Hennigar et al. 2013). While both silviculture and salvage strategies contribute meaningfully to spruce budworm management, such forestry practices alone appear insufficient to prevent outbreak rise or fully mitigate the impacts of outbreaks (Sharov et al. 2002). Consequently, management programs have traditionally depended on large-scale insecticide applications to limit spruce budworm damage (Heartz et al. 2023).

Intensive insecticide-based management of spruce budworm began in the 1950s with the advent of inexpensive and widely available chemical insecticides (Johns et al. 2019; Heartz et al. 2023). Although the original broad-spectrum compounds—such as DDT and fenitrothion—were eventually replaced by more selective bioinsecticides like *Bacillus thuringiensis* var. *kurstaki* (*Btk*) and tebufenozide, the core “Foliage Protection Strategy” (FPS) has remained largely unchanged. FPS involves the targeted application of insecticides to suppress high-density budworm populations, with the goal of keeping defoliation below roughly 50% to minimize tree mortality and growth loss (Fuentealba et al. 2019a). Treatments typically focus on high-value or high-risk stands, especially those dominated by balsam fir, with *Btk* often applied twice in a season—ideally spaced 7 to 15 days apart—to maximize larval mortality. Meanwhile, untreated surrounding forests are left to endure the full outbreak cycle, which eventually collapses on its own. While FPS has proven effective at achieving annual defoliation reduction targets (Fuentealba et al. 2015, 2019a, 2022), treated stands are frequently reinfested from nearby unmanaged areas. As a result, retreatment is often required every one to two years to maintain effective protection (Fuentealba et al. 2019a; Baucé et al. 2024).

In recent decades, advances in ecological theory and management technology have sparked renewed interest in proactive strategies aimed at suppressing insect outbreaks before they escalate and spread (Sharov et al. 2002; Carroll 2007). One such approach for spruce budworm is the Early Intervention Strategy (EIS), which has been increasingly described in recent literature, including its empirical foundation (Régnière et al. 2019), conceptual framework (Johns et al. 2019), and operational implementation (MacLean et al. 2019). In essence, EIS is an adaptive area-wide management program designed to prevent the rise and spread of outbreaks by targeting low-density but increasing populations—referred to as “hotspots”. These hotspots are defined as sample sites where the average density of overwintering second-instar larvae (L2) reaches or exceeds 7 L2 per 75 cm branch (Régnière et al. 2019). While this density may cause detectable defoliation, it generally remains below the threshold for significant larval competition or measurable

impacts on tree growth and survival. Although many factors can affect spruce budworm population performance (survival and reproduction) and determine the precise value of this threshold, populations below that threshold are typically constrained by mate-finding difficulties (Régnière et al. 2012) and high levels of natural predation (Bouchard et al. 2018; Régnière et al. 2019), effects collectively known as depensatory pressures or Allee effects (Courchamp et al. 1999; Johns et al. 2019). Once populations exceed this critical density—due to a decrease in local mortality rates, an increase in immigration, or both (Régnière et al. 2019)—they are more likely to grow rapidly and serve as sources of dispersing moths that contribute to outbreak expansion (Régnière and Nealis 2019). In contrast to the FPS, which focuses on safeguarding high-value or high-risk stands, EIS seeks to detect and suppress emerging hotspots regardless of their location. As such, its effectiveness is judged not by its ability to reduce immediate defoliation, but by its success in preventing hotspot growth and the broader spread of outbreaks—though a significant reduction in defoliation often follows as a secondary benefit.

In this article, we present findings from a 10-year case study evaluating the effectiveness of an operational EIS program for managing a spruce budworm outbreak in New Brunswick, Canada. We assessed the program's effectiveness at two scales, each representing a different level of impact and return on investment. First, we evaluated its success in suppressing local population growth within areas treated with insecticides—what we refer to as “hotspot suppression”. Second, we examined whether this localized hotspot suppression translated into broader, “area-wide suppression”, by limiting overall population growth and spread across the managed region in northern New Brunswick. To provide experimental controls, we compared population data from untreated areas immediately adjacent to the treatment zones, as well as from Quebec's Lower St. Lawrence and Gaspé regions, where a concurrent spruce budworm outbreak was underway (Fig. 1). In Quebec, the outbreak was managed using the traditional FPS, enabling us to highlight key differences between EIS and FPS in terms of insecticide use and their respective impacts on outbreak dynamics (i.e., how populations rise to destructive levels and spread). However, it is important to note that our study does not constitute a direct comparison of EIS and FPS effectiveness. Rather, it focuses on the distinct ways each strategy influences outbreaks. While FPS may use some of the same tactics as EIS, it is designed to limit tree mortality, not target hotspots or alter the course of an outbreak. This distinction makes FPS-managed areas useful controls for evaluating EIS, as any differences in outbreak trends can more confidently be attributed to the proactive nature of EIS itself.

Methods

Spruce budworm life cycle

Spruce budworm is a univoltine species (Swaine 1924) that overwinters as an L2 and emerges in the spring, slightly ahead (1–2 weeks) of the swelling buds of its host. Before budburst, larvae mine mature needles for nourishment, but as the bud cap begins to thin, they quickly move into the swelling buds (Trier and Mattson 1997). Once new shoots elongate, the larvae construct loose silken tunnels, often between shoots, where they continue feeding until they reach the sixth instar (L6). By mid-June to late July, depending on climate, the larvae pupate, emerging as moths 10–15 days later. Moths mate and disperse during the reproduction period (Rhainds and Kettela 2013), with females laying egg masses on individual needles in batches of approximately 5–30 eggs. These eggs hatch within two weeks, and the hatchlings disperse either locally by crawling or by ballooning on an extruded silken thread. Once the first-instar larvae settle on a suitable site, they crawl beneath bud scales, male cone bracts, or furrows on branches and trunks, where they spin a hibernaculum. Here, they molt to L2 and enter obligate diapause to hibernate through the winter (Jaynes and Speers 1949). In the spring, the larvae emerge once again, dispersing to new feeding sites and beginning the cycle anew.

Data sources

We used spruce budworm L2 abundance and aerial defoliation survey data to evaluate EIS effectiveness. We specifically examined sites where hotspots (i.e., ≥ 7 L2 per 75 cm branch) were consistently detected for at least 3 consecutive years, which for the Lower St-Lawrence and Gaspé regions of Quebec included 2010–2024 (Ministère des Ressources Naturelles et des Forêts), and for New Brunswick included 2014–2024 (New Brunswick Department of Natural Resources and Energy Development) (Table 1; Fig. 1). In Quebec, three hotspots observed in 2007 were not included as they did not reoccur in the following two consecutive years (2008 and 2009). Hotspots were detected again in 2010 ($n=6$) and were found consistently thereafter. Both FPS and EIS used the same sampling methodology to monitor overwintering L2. In brief, sampling occurred in the fall after the L2 had settled in hibernacula for overwintering. Sites were selected mainly along roadways and technical staff collected a 75 cm mid-crown branch tip from each of three dominant or codominant hosts, which were mainly balsam fir and white spruce (*Picea glauca* [Moench] Voss), and occasionally

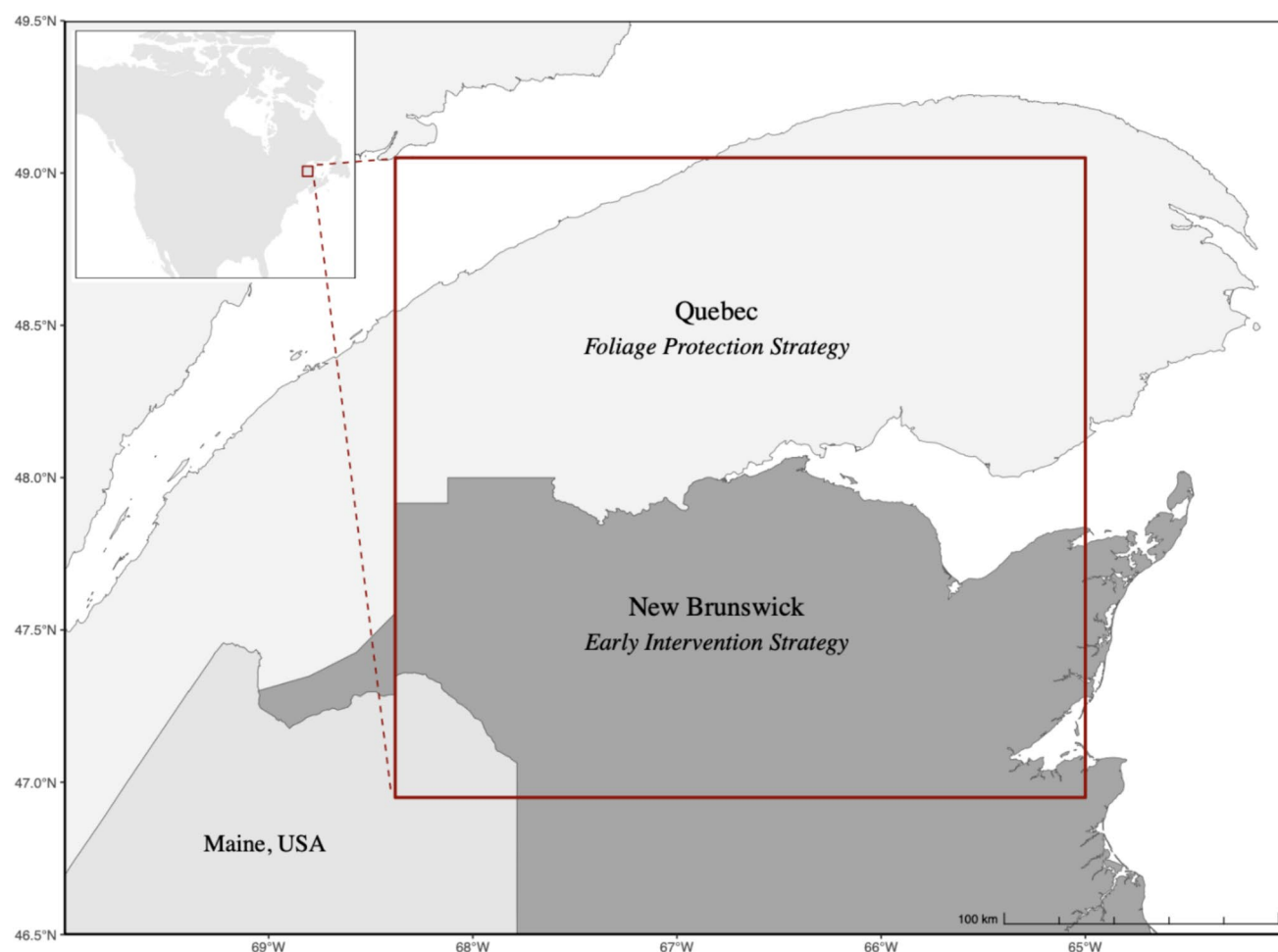


Fig. 1 Map of study areas where Early Intervention Strategy (New Brunswick—dark gray) or Foliage Protection Strategy (Quebec—light gray) were used to manage spruce budworm in eastern Canada

from 2014–2024. The boundary of our study area was set to capture >95% of treatments applied in New Brunswick and produced approximately equal land areas for each program

Table 1 Comparisons of the two insecticide-based management strategies for spruce budworm, the Early Intervention Strategy and the Foliage Protection Strategy. Estimates reflect the areas defined in Fig. 1 covering our total study period from 2010–2024

Strategy	Early Intervention Strategy (EIS)	Foliage Protection Strategy (FPS)
Aim	Suppress hotspots with the aim of suppressing outbreak rise and spread	Suppress defoliation below 50% in high-risk stands to limit tree mortality
Action threshold	Local population density ≥ 7 L2 per 75 cm branch (i.e., ‘hotspot’) in stands with $\geq 10\%$ softwood content (i.e., balsam fir and spruce)	A single year of moderate-severe defoliation in a ‘high-risk’ or ‘high priority’ forest stand (i.e., mainly mature balsam fir and white spruce)
Province	New Brunswick	Quebec
Study years ^a	2014–2024	2010–2024
Treatment years	2015–2024	2014–2024
L2 sample plots per year (range; average)	1212–1423 (avg. 1279)	86–2082 (avg. 957)
Percent total area per treatment regimen ^b	Single Btk (43%) Single Tebufenozide (47%) Double Btk (10%)	Single Btk (29%) Double Btk (71%)

^aThe first year listed was the first year when at least one end-of season hotspot was detected and found consistently thereafter

^bSecond applications were generally applied within 1–2 weeks after the first application

black (*Picea mariana* [Mills] B.S.P.) or red spruce (*Picea rubens* Sarg.). For FPS, additional L2 survey data collected by Société de Protection des Forêts contre les Insectes et Maladies (SOPFIM) were included to bolster sample sizes in and around treatment blocks. Collected branches were bagged and transported to regional laboratories where the L2 were extracted using an NaOH wash technique that dissolves the hibernaculum to dislodge the L2 from their overwintering sites (Miller, 1958). Extracted L2 were sifted from the larger branch debris onto gridded filter paper and counted under a dissecting microscope. Data from the three branches were averaged for each site and used to identify hotspots. SOPFIM L2 samples collected from balsam fir trees often had truncated counts (i.e. because the densities were often extreme counting stopped once number on a branch reached 50 L2). A priori analyses of data from Ministère des Ressources Naturelles et des Forêts (i.e., non-truncated counts) revealed a strong correlation between L2 densities on balsam fir and white spruce when both were sampled in the same stands (Pearson's $r = 0.78$). Truncated counts could not be used for estimating population densities and growth rates and given the high correlation between balsam fir and white spruce, we omitted L2 densities collected from balsam fir from the SOPFIM data when estimating these parameters.

Annual defoliation associated with spruce budworm was assessed using a combination of aerial surveys and Landsat Sentinel satellite imagery, the details and history of which have been described for Quebec (Ministère des Ressources Naturelles et des Forêts) and New Brunswick (Summary of Forest Pest Conditions 2021). In general, these methods are most effective for detecting moderate to severe defoliation covering relatively large areas, with the satellite imagery approach being slightly more effective at detecting small patches.

Insecticide treatments

General approaches to treatment block prioritization are discussed for EIS by MacLean et al. (2019) and for FPS by Fuentealba et al. (2022). In brief, EIS identifies and tracks hotspots detected by annual sampling of overwintering L2, forming the initial basis for prioritizing areas for treatments (Johns et al. 2019; MacLean et al. 2019). The occurrence of hotspots was used in conjunction with forest composition to develop a treatment priority layer, which in turn provided the basis for planning treatment areas and flight trajectories for insecticide applications (MacLean et al. 2019). FPS, in contrast, initiates treatments in designated high-priority or high-risk stands after the first year of moderate-severe defoliation and continues annual treatments every 1–3 years as needed until the outbreak collapses in the area (Fuentealba et al. 2019a, b). However, because of the large-scale nature

of outbreaks and high treatment costs, contemporary FPS is applied to only a subset of stands subjected to moderate-severe defoliation (Chang et al. 2012a; Fuentealba et al. 2019a; Bause et al. 2024).

There were two insecticides used for controlling spruce budworm, both of which were mainly applied via fixed-wing aircraft. Both EIS and FPS used the bacterial insecticide *Btk*, which needs to be ingested and is relatively specific to larval lepidopterans. Another insecticide only used in EIS was the synthetic moulting hormone analog tebufenozide (CAS: 112,410–23-8), which, similar to *Btk*, needs to be ingested and primarily impacts lepidopterans. In general, single applications of *Btk* and tebufenozide were used in EIS, although ~10% of treated area received a double application of *Btk* (Table 1). In FPS, only *Btk* was used, but ~71% of treated areas received a double application of *Btk* with the remainder receiving only one treatment per year (Table 1).

Study area

In New Brunswick, we focused on the northern region, which is where most hotspots occurred and all EIS treatments applied (Fig. 1). Two small treatment blocks just south of this area in 2021 and 2022 (16.5 km² and 2.8 km² respectively; <0.5% of the total area treated under EIS) were excluded. Removing these blocks allowed us to compare two approximately equal study areas by extending the selection box northward into Quebec (25,000 and 26,000 km² in New Brunswick and Quebec respectively; Fig. 1). Within the study boundaries we identified areas that were treated or not treated with insecticides.

Hotspot suppression

We compared hotspot suppression between the EIS and FPS areas, using the FPS as a de facto control since by design it would not be expected to suppress hotspots. In comparing these jurisdictions, we assessed annual population growth rates, end-of-season L2 densities, and the percent change in the number of hotspots in treated and untreated areas in each of the management jurisdictions within our study area (Fig. 1). Untreated areas were restricted to be within 5 km of treated areas to limit potential differences in site characteristics and moth immigration pressure while still providing sufficient L2 sampling points for comparison. Often L2 sampling did not occur at the exact same locations among years, so samples were grouped into sites by identifying clusters of points within 1 km of each other. L2 densities were averaged if multiple samples were collected within our site designation for a given year. Population growth rate (R) per site was estimated as: $R = \log(L2_t/L2_{t-1})$, where $L2_t$ is the end-of-season density in year t and $L2_{t-1}$ is the pre-season density. For each program growth rates were analyzed

with a linear model with fixed effects of year, treatment, and their interaction and pre-season L2 density included as a covariate. End-of-season L2 density was analyzed using a linear model fit with fixed effects of year, treatment and their interaction; density was log transformed prior to analysis. To assess differences in the occurrence of hotspots before and after treatments were applied, we modeled the change in the percent of sites classified as hotspots from pre- to end-of-season using a linear model with fixed effects of year and treatment. All analyses were run using R version 4.4.2 (R Core Team 2024).

Area-wide suppression

To demonstrate the extent to which EIS suppressed populations right up to the border between New Brunswick and Quebec ($\sim 48^\circ\text{N}$; Fig. 1) we first overlaid a grid over our study area. Grid spaces were set to hexagons with 20 km between opposite sides, each side was ~ 11.5 km long creating a total area of 346 km^2 . L2 densities were averaged per grid cell per year. Differences in L2 density along a latitudinal gradient (taken from the center of each grid cell) during the duration of EIS (2015–2024) were plotted to show rise and spread across the landscape.

Since hotspots first appeared at different times in each province (i.e., 4 years earlier in Quebec), to examine area wide suppression comparisons were made starting from when hotspots were consistently detected for at least three years in provincial surveys (i.e., “year 0”), which first occurred for FPS in 2010 and for EIS in 2014. Note that L2 values are overwintering end-of-season estimates, therefore hotspots were first detected in the fall of year 0 and treatments applied in the spring of the following year (e.g. year 1).

To examine overall trends in population growth rates for each treatment program, we estimated population growth rates for all sites within our study area starting at year 1. Because population growth can be influenced by larval densities, growth rates were expressed as a function of their pre-season ($L2_{t-1}$) densities. For simplicity, pre-season L2 values were grouped into 15 equally spaced bands on the log scale. Weighted polynomial linear models were used to assess the relationship between growth rate and pre-season L2 density (per program) for treated and untreated areas. Models included treatment status (treated or untreated) as a fixed effect and $\log(L2_{t-1})$ as a covariate and were weighted by sample size. Models were fit using polynomial degrees from 1–4 and AIC values were used to select the best fit for each program.

To show how L2 density and hotspots progressed under each program among years we grouped estimates of L2 density starting at year 0 for each program into a hexagonal grid (using methods described above). For each grid space

we reported average L2 density and if hotspots were present (binary yes/no). Average L2 end-of-season density and percent of grid spaces with hotspots were reported for each year. Additionally, we calculated the total area of moderate to high defoliation observed from aerial surveys annually.

To quantify insecticidal use under each strategy, we first calculated the total number of hectares treated with insecticides per year (starting in year 1). Annual treatment blocks were then rasterized using the *rast* function (resolution = $0.0001 \cong 82\text{ m}^2$ cells) from the *terra* package (v.1–78; Hijmans 2024) in R (R Core Team 2024). These rasters were overlaid to calculate the annual frequency of insecticide application per cell. To account for cases where double applications occurred within the same year, application frequency was also reported by counting double-treated areas as two applications rather than one.

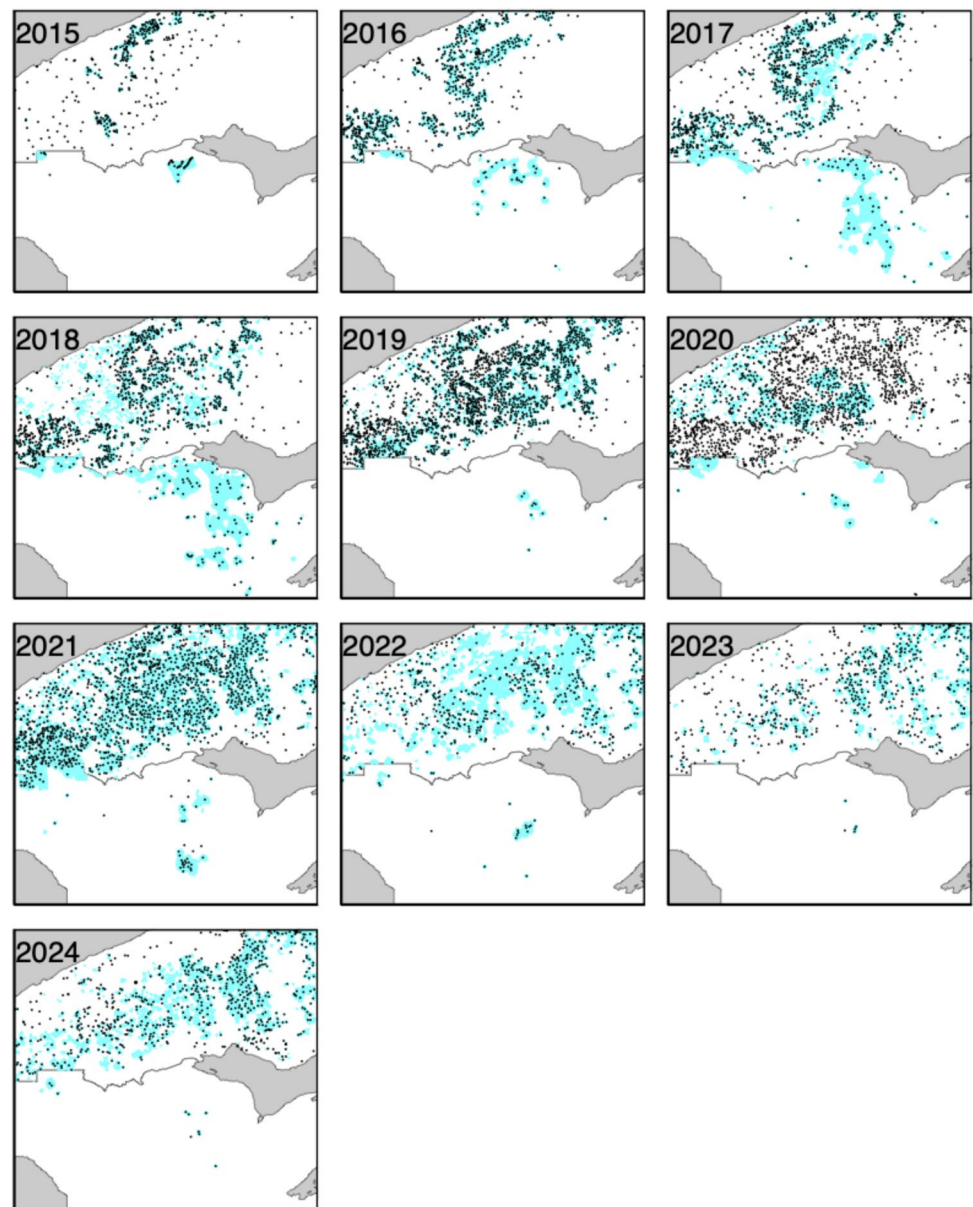
Results

Hotspot suppression

Over the course of our study, hotspots progressively spread to cover most of the FPS area in Quebec, which was expected given that FPS is not designed to control hotspots (Régnière et al. 2019, see also Régnière and Nealis 2019). In contrast, they generally remained few and clustered in two main portions of the EIS area in the northwest and east (Fig. 2). Insecticide treatments in EIS covered an average of 76% (range ~ 70 –90%) of detected hotspots each year whereas FPS treated an average of 44% (range ~ 20 –85%) of hotspots each year. Untreated hotspots under EIS reflected areas where treatments were restricted by regulations, such as water buffers, habitations, and windmill farms.

In the EIS-treated area there were significant differences in population growth rates between treated and untreated areas (Table 2). In all years, population growth rates in EIS-treated areas were considerably lower than in surrounding untreated areas (Fig. 3a). Average L2 densities in treated areas at the end of the treatment season were significantly different from those in untreated areas (Table 2), treated areas tended to be slightly lower than untreated (Fig. 3b) but there was a significant interaction of treatment effects among years. Most importantly, in most years (2015–2023), mean post-treatment L2 density was below the EIS action threshold (0.41 – 3.72 L2 per branch; Fig. 3b) which was the objective. EIS significantly reduced the number of hotspots in treated areas by 12–68% (Fig. 3c; Table 2). Inside the treatment areas, 16–67% of sample sites were hotspots pre-treatment, and apart from 2024 0–17% post-treatment. In 2024 47% of post-treatment samples from inside treated areas were hotspots. In nearby untreated areas, the number of hotspots increased in most years by 4–27% (Fig. 3c). For

Fig. 2 Hotspots (pre-treatment) and insecticide treatment maps for areas managed via the Early Intervention Strategy (New Brunswick) and the Foliage Protection Strategy (Quebec) from 2015 to 2024. Hotspots (i.e., sites with ≥ 7 second-instar larvae per 75 cm branch) are marked (circles) and treatment areas are coloured (cyan). Note that from a management standpoint hotspots reflect an action threshold relevant to EIS, but not FPS



FPS, treated areas had negative population growth rates in about half of the years, though the magnitude of the effect was lower than for EIS (Fig. 3d). There was a significant difference in population growth rates between treated vs. untreated areas. However, this relationship is not straightforward, as there was also a significant treatment $\times$ year interaction (Table 2). FPS showed no significant main effect of treatment on end-of-season L2 densities, but did suggest a significant year by treatment interaction indicating that no consistent overall differences were found between treated and untreated areas across years (Table 2). Overall, FPS treated areas had end-of-year densities between 8.75–58.1 L2 per branch (Fig. 3e). There was a significant difference in the percent of sites that were hotspots in treated and untreated areas (Table 2). FPS treatments reduced the

percentage of sites that were hotspots by 0–52% (Fig. 3f). Inside the treatment areas, 85–100% of samples were hotspots pre-treatment compared with 35–99% post-treatment, while in nearby untreated areas hotspots generally increased by 0–40% (Fig. 3f).

Area-wide suppression

Since the first hotspots were detected in New Brunswick (2021), there was a clear separation in pre-season L2 density ranges at the provincial border (48°N) separating EIS (New Brunswick) and FPS (Quebec) managed areas (Fig. 4). EIS successfully suppressed hotspots despite the high density

Table 2 Results of linear models testing effects of year, treatment, pre-season L2 densities and their interactions on spruce budworm population growth rates, end-of season L2 densities, and the difference in the percentage of hotspots under the Early Intervention Strategy (EIS) and the Foliage Protection Strategy (FPS)

Measure	Program	Factor	DF	MS	F	<i>p</i>
Population growth	EIS	Year	9	28.72	76.65	< 0.001
		Treatment	1	194.33	518.67	< 0.001
		Pre-season L2	1	104.53	278.99	< 0.001
		Year:Treatment	9	0.68	1.81	0.062
		Residuals	1481	0.38		
	FPS	Year	9	29.18	151.63	< 0.001
		Treatment	1	34.94	181.53	< 0.001
		Pre-season L2	1	75.93	394.56	< 0.001
		Year:Treatment	9	0.81	4.23	< 0.001
		Residuals	1555	0.19		
L2 density (end-of-season)	EIS	Year	9	54.31	59.42	< 0.001
		Treatment	1	20.09	21.98	< 0.001
		Year:Treatment	9	4.16	4.55	< 0.001
		Residuals	2325	0.91		
	FPS	Year	9	233.89	229.10	< 0.001
		Treatment	1	2.09	2.04	0.15
		Year:Treatment	9	7.70	7.54	< 0.001
		Residuals	2991	3053.60	1.02	
	EIS	Year	9	0.03	2.70	0.08
		Treatment	1	1.08	98.69	< 0.001
		Residuals	9	0.01		
	FPS	Year	9	0.08	6.91	0.004
		Treatment	1	0.26	21.48	0.001
		Residuals	9	0.01		

Degrees of freedom (DF), mean squares (MS), F statistics and associated *p*-values are reported; significant *p*-values are in bold

of L2 immediately across the border into Quebec where the FPS program was ongoing (Fig. 4).

When looking at all years since hotspots were consistently detected, in both programs population growth rates declined with increasing pre-season L2 density at a greater rate within treated areas (Fig. 5). Model selection indicated that a simple linear model was the best fit for the EIS data and a third-degree polynomial was the best fit for FPS. For both programs population growth rates were significantly related to pre-season L2 density (EIS: $p < 0.001$, $F_{1/20} = 817$ and FPS: $p < 0.001$, $F_{3/19} = 62$) and were significantly lower in treated areas (EIS: $p < 0.001$, $F_{1/20} = 381$ and FPS: $p < 0.001$, $F_{1/19} = 153$).

For the first three years following the consistent detection of hotspots, average L2 density increased at a lower rate under EIS than FPS (Fig. 6a), whereas the percentage of our study areas (i.e., Fig. 1) with hotspots under the two strategies increased at a similar rate (Fig. 6b). After this initial rising phase, however, both L2 density and percent area with hotspots continued to rise under FPS, but dropped to very low levels under EIS. Although populations fluctuated in the FPS area, the proportion of hexagons containing hotspots continuously increased, reaching and remaining over

88% between years 6 and 14 (Fig. 6b). The area suffering moderate to severe defoliation (Fig. 5c) was far more extensive under FPS, peaking at ~1.5 million ha in year 7 (2017) compared with the peak under EIS that reached ~65,000 ha in year 7 (2021).

The total area treated with insecticides under EIS increased in tandem with the occurrence of hotspots, peaking in year 4 (2018) at ~198 k ha of forest which amounted to ~220 k ha total treated area after accounting for areas where double insecticide applications were done. After this initial peak, treatment area declined to ~10 k ha in the following year (2019) (Fig. 7a). A second, minor peak occurred in year 7 (2021) when 72 k ha of forest received treatment. The area treated again declined (~14 k ha) in the following year (Fig. 7a). In contrast, insecticide treatments under FPS started later, five years after the first occurrence of hotspots, and increased to a peak of ~433 k ha of forest selected for treatment in year 11 (Fig. 7a). Accounting for double pesticide applications, this amounted to ~852 ha treated in a single year. During the study period for EIS, most treatment area (~66%) was treated only one year (Fig. 7b) and 58% was treated with a single application of pesticide (Fig. 7c). Less than 5% of

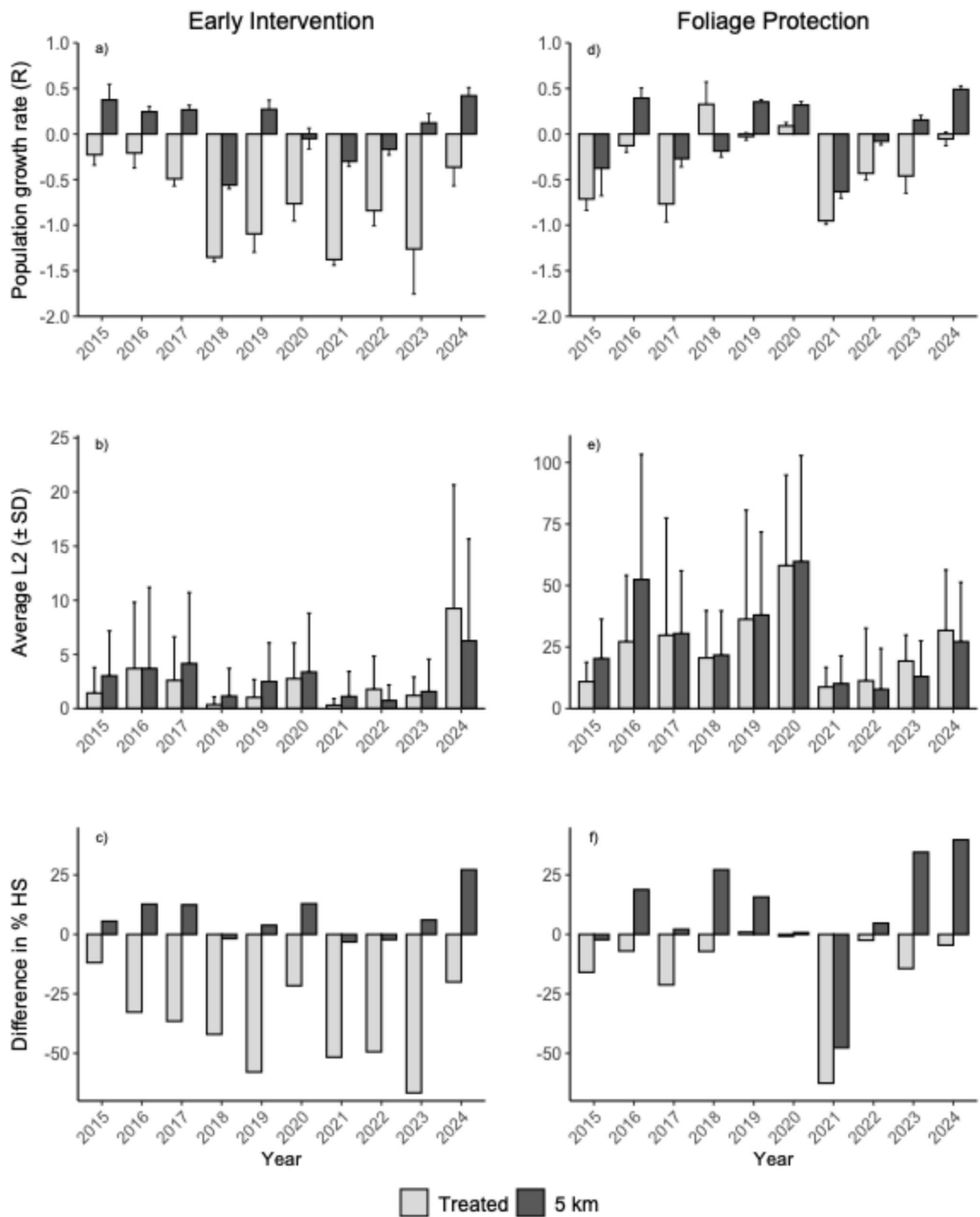


Fig. 3 Impact of Early Intervention Strategy and Foliage Protection Strategy on (a, d) population growth rates, (b, e) end-of-season second-instar larvae densities ($L2 \pm 1SD$), and (c, f) percent difference in

hotspots in treated vs. untreated areas from 2015–2024. Estimates for untreated areas were restricted to within 5 km of the treatment block edge. Note the difference of scale on the y-axis in panels b and e

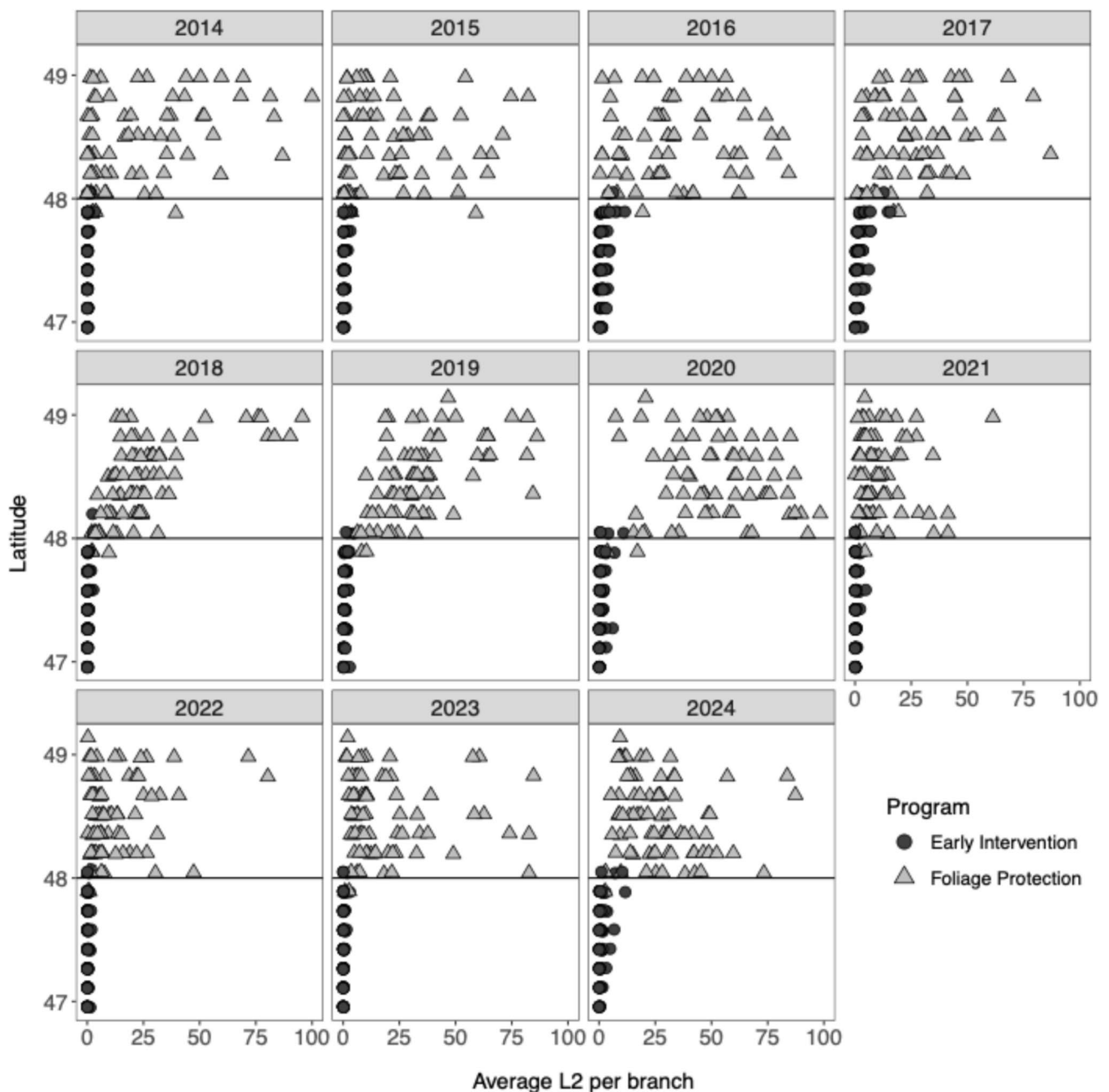


Fig. 4 Average spruce budworm density along a latitudinal gradient from the northern to southern range of our study areas for Early Intervention Strategy (EIS) and Foliage Protection Strategy (FPS). Densities represent the mean second-instar larvae (L2) density calculated for each hexagonal grid area. Grid spaces were set to 20 km across.

A few areas in the FPS treatment area had average L2 densities > 100 per branch but are not shown for simplicity. Latitude of 48°, shown by the horizontal line, is the approximate political boundary separating EIS and FPS treated areas

the treated area was treated in four years or more (Fig. 7b), or with 4 or more pesticide applications (Fig. 7c). In contrast, over the 14-year FPS study period, over 50% of the total treatment area was treated in 4 or more years, with a

small number of areas being treated up to 8 years (Fig. 7b). Taking annual double applications into account, ~55% of the treated area received 6 or more insecticide applications

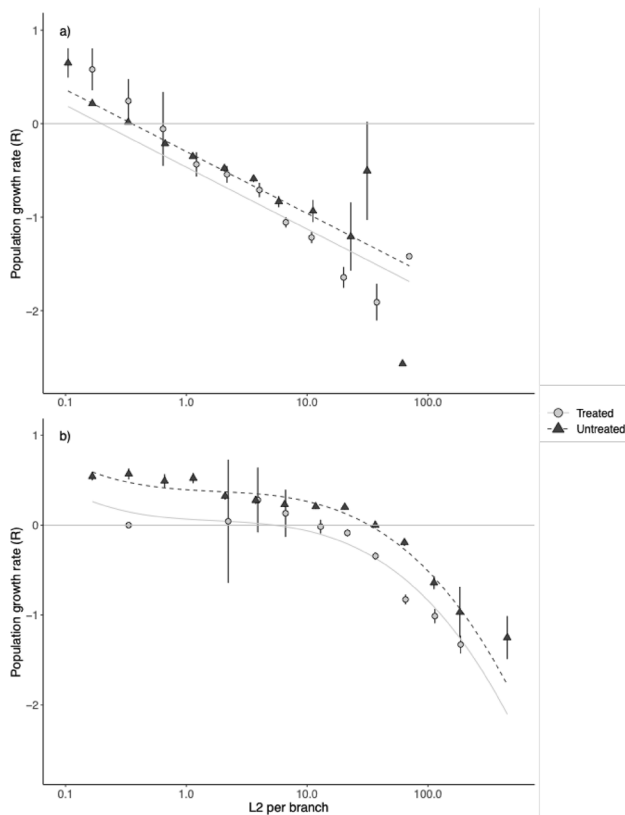


Fig. 5 Population growth rates ($R \pm 1$ SE) of spruce budworm in treated and untreated areas as a function of pre-season second-instar larvae (L2) density, under the (a) Early Intervention Strategy and (b) Foliage Protection Strategy. Growth rates were pooled across years starting from the year immediately after consistent hotspots were detected (2015–2024 for EIS; 2011–2024 for FPS). Note that the x-axis is presented on the log scale

(maximum 14) and only 11% received a single application (Fig. 7c).

Discussion

Our study provides strong evidence of the high effectiveness of EIS in suppressing local spruce budworm outbreaks. In all years, EIS insecticide applications led to significant suppression of hotspots in treated areas (Figs. 2, and 3c), with population growth rates and the number of hotspots consistently much lower than those in neighboring untreated forests (Fig. 3a). Across the entire EIS-managed area, populations grew for a few years after the initial detection of hotspots but declined sharply in 2018 and remained low thereafter, indicating that EIS also provided sustained area-wide suppression within the management zone (Figs. 4 and 6a, b). As a result, in contrast to the FPS area, defoliation was negligible

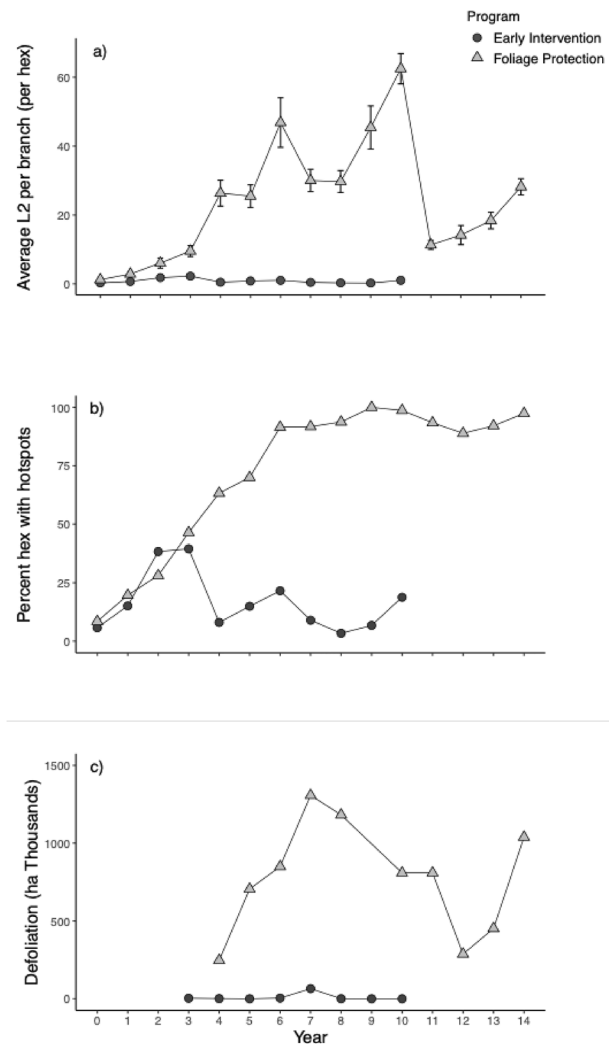


Fig. 6 Progression of spruce budworm outbreak and management associated with Early Intervention Strategy and Foliage Protection Strategy per year, beginning the year that the first hotspot was detected in the two programs: a average second-instar larvae (L2) per branch per grid area, b percentage of grid areas with hotspots, and c number of hectares sustaining moderate to severe defoliation. Plot densities were averaged for hexagonal areas (20 km across). The first year a hotspot was detected (Year “0”) in surveys for Foliage Protection Strategy was 2010 and for Early Intervention Strategy was 2014

throughout the study period in the EIS area (Fig. 6d). The effectiveness of EIS for imposing area-wide suppression of hotspots was particularly striking when compared to the population trends to the north, managed by FPS, which were seemingly unaffected by insecticide treatments. This resulted in a more typical outbreak “cycle,” with large areas experiencing moderate to severe defoliation (Fig. 6a, d). While hotspots did recur in some regions of EIS-managed areas—such as the northwest corner of New Brunswick, near the Quebec border, and the north-central portion south of Campbellton, NB—relatively small to modest treatment

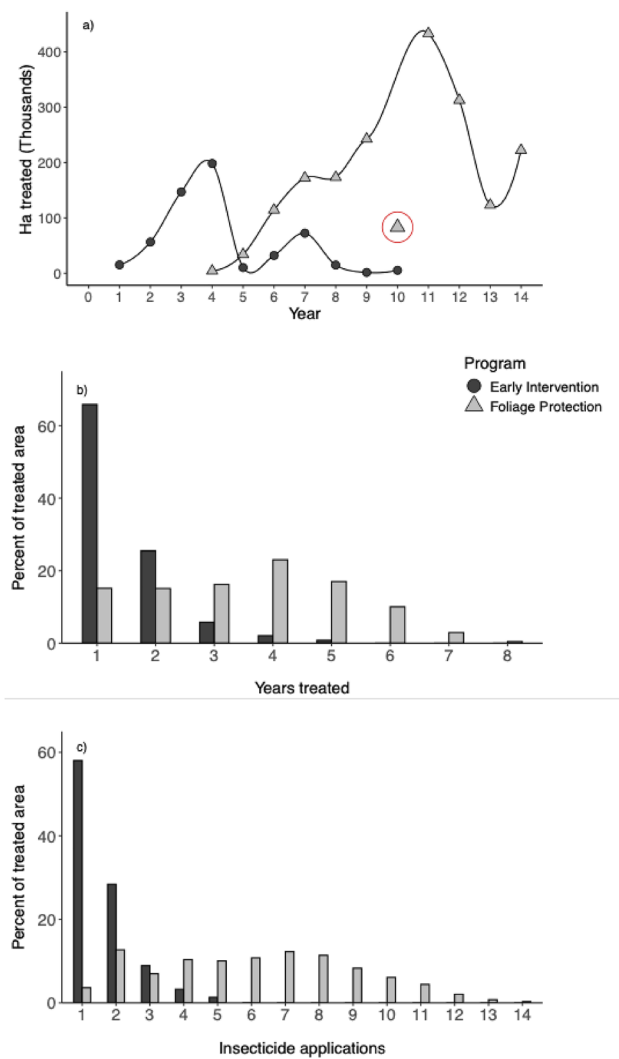


Fig. 7 Insecticide treatment operations under Early Intervention Strategy and Foliage Protection Strategy: **a** area (ha) treated annually with insecticides, and **(b)** percentage of total treated area that received treatments in multiple years, and **c** percentage of total treated area that received more than one insecticide application, accounting for both repeated applications among years and within in the same year (Tabel 1). The circled point in **(a)** indicates a year where the extent of areas meant to be treated were reduced due to the Covid-19 pandemic restrictions (2020)

areas targeting hotspots were sufficient to prevent the establishment and spread of the outbreak. Overall, both EIS and FPS influenced outbreak trends in ways that aligned with their respective goals, designs, and underlying ecological theory (Pureswaran et al. 2016; Johns et al. 2019; Cooke and Régnière 2025).

As predicted, EIS had a significantly greater impact on suppressing population growth and associated hotspots compared to FPS (Fig. 2, and 3a, d), despite relying primarily on single insecticide applications (Table 1). This is likely because EIS prevented the density-dependent compensatory

responses that can allow populations to rebound under FPS. While FPS aims to reduce populations and preserve foliage, this approach is likely to alleviate mortality caused by larval competition for food in treated areas, leading to population growth rates more or less matching those in untreated areas (Fig. 3d). Additionally, because FPS targets mostly high-value stands, budworm from surrounding untreated forests can readily reinvade treated areas, counteracting any population reductions achieved through treatment. Our study was unable to distinguish the relative contribution of these compensatory feedbacks. Overall, however, population growth rates and L2 densities remained high in FPS-treated areas (Fig. 3d, e), and suppression was ostensibly short-lived given the high frequency of retreatment needed in most areas (Fig. 7b). This aligns with past studies on FPS, which highlight the need for retreatment every 1–2 years to maintain protection (Fuentealba et al. 2019a; Baucé et al. 2024). Overall, our EIS results suggest that compensatory feedbacks (i.e., competition for food or reinvansion) were not strong enough to offset the mortality caused by insecticide applications in any year.

Assessments of population growth rates across varying densities indicate that EIS treatments are most effective at high population densities and less so at lower densities (Fig. 5a). This pattern likely reflects density-dependent moth dispersal dynamics (Régnière and Nealis 2019). Typically, high-density populations experience net dispersal losses as individuals move into surrounding lower-density areas, while low-density populations gain individuals from these sources. Under EIS, however, treated high-density populations are effectively isolated—surrounded by low density or similarly treated areas—so mortality was not offset by immigration. In contrast, at lower densities, treatment-induced losses may be partially compensated by modest immigration from surrounding, moderately dense populations or from the outbreak to the north in Quebec. This result suggests that EIS dilutes populations across the managed landscape, reducing both local growth rates and average regional densities. Under FPS, by comparison, reductions in population growth at high densities are constrained by reinvansion from nearby untreated areas where populations remain high. Moreover, the observed curvilinear reduction in growth rates at very high densities may reflect intraspecific competition or increased emigration driven by foliage depletion (Fig. 5b). Notably, the density threshold at which growth rates decline sharply under FPS (~10–30 L2 per branch, Fig. 5b) coincides with densities where intraspecific competition would be expected to intensify (Régnière et al. 2019). When competitive pressures or emigration are alleviated by FPS treatments, any potential reductions in annual population growth rates may be further diminished. These findings underscore the importance of maintaining low population densities over

large contiguous areas in order to disrupt dispersal dynamics and achieve the sustained suppression observed under EIS.

Our results suggest that area-wide hotspot suppression was a key factor in limiting the rise and spread of outbreaks in EIS-treated areas across northern New Brunswick, primarily by preventing the establishment and expansion of hotspots. This area-wide suppression was particularly noteworthy for its sustained success, extending all the way to the Quebec border, where hotspots were otherwise common—especially in the northwest (Figs. 2 and 4). This success carries several important implications. From an outbreak dynamics perspective, the ability of EIS to control hotspots and suppress the broader outbreak up to the Quebec border indicates that dispersal from Quebec was not overwhelming during our study period, apart from occasional mass dispersal events that triggered temporary pulses in population growth (e.g., 2017) (Johns et al. 2019). While such mass dispersal events do present a potential challenge, our findings suggest they were relatively infrequent in New Brunswick and had only a limited, manageable impact on hotspot formation and growth. From an EIS management perspective, these results indicate that proximity to a high-density outbreak area does not necessarily lead to uncontrollable outbreaks, providing that the local populations are kept relatively low. This insight is particularly relevant when considering how protected or sensitive areas (e.g., Protected Natural Areas (PNAs), provincial and federal parks, First Nations lands, etc.) might affect EIS outcomes if outbreaks within these zones are allowed to follow their natural course. While our study did not directly address this question—since PNAs in New Brunswick were included in EIS treatments and no major parks experienced hotspots—future insights may emerge from ongoing EIS efforts in western Newfoundland. There, Gros Morne National Park (~ 72,000 ha of forested area) was excluded from insecticide applications, allowing the outbreak to progress naturally within park boundaries.

While the evidence supporting the effectiveness of EIS is compelling, it remains uncertain whether the suppression of outbreaks observed in northern New Brunswick was the reason no hotspots occurred in southern half of the province (i.e., south of the area shown in Fig. 1), or whether other factors are at play. Two primary hypotheses posit explanations for why southern New Brunswick may be less hospitable to spruce budworm compared to previous outbreaks. The first hypothesis invokes climate change. There is clear evidence that climate change has driven a significant northward expansion of the spruce budworm's range since the previous outbreak (Liu et al. 2022; Boulanger et al. 2025). It has been suggested that the southern range limit of spruce budworm may have correspondingly shifted northward due in particular to climate-driven changes in overwintering success (Han and Bause 1997, 1998; Roe et al. 2024) and altered synchrony between larval emergence and host tree phenology

(Fuentealba et al. 2019b; Bellemin-Noël et al. 2021). However, experimental results on the potential southern outbreak range are mixed. Laboratory studies report reduced overwintering survival under warmer conditions (Han and Bause 1997, 1998; Roe et al. 2024), while field studies have generally observed only modest impacts, mainly on metabolite concentrations rather than survival rates (Régnière and Duval 1998; Moise et al. 2023, 2024).

The second hypothesis centers on changes in forest composition, which numerous studies have shown to influence outbreak intensity and duration. Since the previous outbreak, New Brunswick's forests have become less susceptible to spruce budworm due to a decline in balsam fir, increases in white and black spruce, a younger forest age structure, and greater proportions of intolerant hardwoods and mixed-wood stands. These changes are especially pronounced in southern New Brunswick, where widespread damage from the previous outbreak prompted a modest shift toward more mixed-wood and hardwood forests (MacLean, 1984). Both hypotheses warrant further investigation, however, the extent to which they contribute to the current success of EIS remains unclear. Notably, while no hotspots have been detected in southern New Brunswick, the outbreak in southwestern Quebec continues to spread southward. As of 2024, patches of severe defoliation have been observed south of 46°N (Ministère des Ressources Naturelles et des Forêts 2024), which corresponds to south-central New Brunswick—approximately 100 km south of the southernmost EIS boundary shown in Fig. 1.

While effectiveness is critical, choosing a management strategy ultimately involves weighing intervention costs against the value of avoided damage, which for spruce budworm relates to tree mortality and growth loss. Numerous studies have demonstrated the high effectiveness of FPS in reducing spruce budworm defoliation (Fuentealba et al. 2015, 2019a, 2021) and its associated economic benefits (Hennigar et al. 2013; a et al. 2012). Chang et al. (2012a) estimated that FPS in New Brunswick could have reduced economic losses by up to 66% by treating just 40% of susceptible forests—an effect that could be further enhanced by an additional 1–18% through strategic harvest rescheduling and timely salvage of dead trees. However, MacLean et al. (2019) noted that the cost of insecticide applications has increased by 10 to 20 times since the last major outbreak, rendering large-scale programs—such as those covering ~ 2 million hectares annually between 1970 and 1983—potentially cost-prohibitive today. Our current study shows that EIS used substantially less insecticide than FPS over comparable periods and required far fewer retreatments (Table 1, Fig. 7). These results suggest that, under the right conditions, EIS could be even more cost-effective than previously estimated by Liu et al. (2020). Ultimately, this comparison underscores the importance of integrating insecticide-based

management with complementary strategies, such as silviculture and salvage, to enhance both ecological and economic outcomes.

The apparent success of EIS in suppressing outbreaks raises an important question: just because we can suppress an outbreak, should that always be the goal? On one hand, there are compelling reasons to prevent outbreaks. Beyond their economic toll, uncontrolled outbreaks can trigger broader ecological impacts—such as intensifying competitive interactions that favor moose over caribou (Labadie et al. 2021) or hindering forest regeneration due to heavy browsing pressure (Leroux et al. 2021). Tree mortality during outbreaks can also reduce forest carbon sequestration capacity (Liu et al. 2022), increase stream discharge (Sidhu et al. 2024), shift algal communities (Ju et al. 2024), and harm cold-water fish species by eliminating the protective shading of riparian overstory. On the other hand, it is critical to recognize that spruce budworm is a native species and a long-standing disturbance agent in the boreal forest, having co-evolved with this ecosystem for millennia (Sturtevant et al. 2015). Like wildfire, budworm outbreaks play a vital role in forest nutrient cycling and structural dynamics. Many native species—including birds, mammals, insects, and plants—depend on the unique habitat features created by defoliation and tree death, including forest gaps and standing deadwood. As such, while suppression may offer clear advantages, it also risks disrupting natural processes that support forest biodiversity and resilience. In regions where the preservation of natural disturbance regimes is a priority, a strategy like FPS may be more appropriate. By focusing protection efforts on high-priority forests while allowing outbreaks to proceed naturally elsewhere, FPS offers a balanced approach that safeguards key assets without wholly suppressing ecologically beneficial dynamics.

From a practical standpoint, EIS may not be feasible in all areas due to several challenges, including the need for extensive road access to support intensive L2 monitoring, dedicated and efficient L2 branch processing facilities, strong and sustained financial and industry backing well before significant damage occurs, and broad sociopolitical acceptance (Johns et al. 2019; MacLean et al. 2019). In addition, EIS relies heavily on active, widespread monitoring to detect emerging hotspots early. However, ongoing monitoring during the long intervals between outbreaks is not typical in Canadian forest pest management, where spruce budworm risks tend to receive more attention during epidemic phases. Maintaining distributed, continuous monitoring will be critical for generating the intelligence needed to guide both EIS and FPS programs. A key takeaway from this work is that proactive—rather

than reactive—monitoring and management could deliver a stronger return on investment.

Our study of the EIS program underscores an important point: the success of a pest management program depends on more than just the larval mortality rate achieved by insecticides. Among all insecticides historically used to control spruce budworm—including potent organophosphates and organochlorines—the current suite of bioinsecticides used in EIS produces the lowest overall larval mortality. For example, historical use of DDT in northern New Brunswick achieved mortality rates of up to ~99% (Webb 1959; Secrest and Thornton 1959), whereas field trials of Btk typically report mortality rates of around 80%, compared to ~66% in untreated sites (Fuentealba et al. 2015). Despite this lower mortality, Btk is particularly well-suited for EIS because of its specificity. It targets only lepidopteran larvae that are actively feeding at the time of application and is short-lived. This selectivity minimizes negative impacts on natural enemies and helps preserve non-target lepidopterans that support important parasitoid populations (Glaus et al. 2023). Furthermore, parasitized larvae—which feed less—are less likely to ingest a lethal dose of Btk, enabling healthy larvae to be removed more effectively while leaving parasitized larvae alive until parasitoids can emerge (Nealis and van Frankenhuyzen 1990). Future improvements to EIS should focus on enhancing the specificity and cost-effectiveness of control tactics, rather than simply pursuing higher mortality rates. Our findings reinforce the principle that how insecticides are used—the strategy behind their application—can be as important, if not more so, than their inherent lethality.

In recent decades, area-wide management programs for insect pests have expanded substantially, particularly within agricultural systems (Koul et al. 2008). In forestry, much of the foundational work on outbreak suppression and containment began with the “Slow the Spread” program targeting the invasive defoliator *Lymantria dispar* in the United States (Sharov et al. 2002). This initiative helped pave the way for similar strategies aimed at native pests, including efforts in western Canada to contain the eastward expansion of mountain pine beetle (Carroll 2007; Johnson and Lewis 2025). These programs highlight several ecological principles critical to successful outbreak suppression—such as the role of Allee thresholds, the need for efficient and accurate monitoring, and the use of narrow-spectrum control tactics that contribute additive mortality (Liebhold and Tobin 2008; Johns et al. 2019, 2025). Their demonstrated success suggests that controlling forest insect outbreaks is not only feasible but also a promising approach for other pest management contexts.

Conclusion

Managing large-scale insect outbreaks is inherently complex, and no single strategy will suit all jurisdictions. In this context, EIS and FPS should not be viewed as mutually exclusive but rather as complementary tools within a forest manager's toolkit. Both approaches have demonstrated effectiveness in achieving their respective goals, each offering distinct advantages, limitations, and risks. The key challenge for managers is to determine which strategy—or combination of strategies—best aligns with their jurisdiction's priorities, capacities, and constraints, including available resources, infrastructure, and ecological conditions. For regions with large management areas, a hybrid approach incorporating both EIS and FPS, along with silviculture and salvage operations, may offer the greatest overall benefit. Looking ahead, research should aim to enhance EIS monitoring and operational efficiency, while also exploring how the strategy can be adapted to other jurisdictions and integrated into broader, landscape-scale Integrated Pest Management programs for spruce budworm.

Author contributions

RJ, SE, and JR conceived and designed research. DC, AM, and LAB organized and oversaw all data collection. SE, EM, and PJ analyzed data. RJ wrote the manuscript and incorporated written contributions from all authors. All authors read and approved the manuscript.

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Data availability The data that support the findings of this study were obtained from annual forest insect monitoring datasets collected and maintained by the New Brunswick Department of Natural Resources and Energy Development and the Ministère des Ressources naturelles et des Forêts du Québec. These data are preserved and available through the respective agencies. Access to the data may be subject to agency policies and permissions.

Declarations

Conflict of interest The authors declare no conflict of interests.

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